MongoDB CRUD Operations – Practical Lab Exercises (Assignment 2)

Subject: Advanced Database Management System

Practical Title: Implementation of CRUD Operations using MongoDB Shell (mongosh)

```
show collections

show dbs

db.students.insertOne({rollno: 101,name: "Rahul",age: 22,course: "MCA",semester: 1,city: "Rajkot",marks: 85})

db.students.insertMany([{rollno:102,name:"Priya",age:23,course:"MCA",semester:1,city:"Ahmedabad",mark:90},
                        {rollno:103,name:"Amit",age:22,course:"MCA",semester:3,city:"Rajkot",mark:75},
                        {rollno:103,name:"Neha",age:24,course:"MCA",semester:3,city:"Surat",mark:88},
                        {rollno:105,name:"Karan",age:21,course:"MCA",semester:1,city:"Jamnagar",mark:65},
                        {rollno:106,name:"Riya",age:23,course:"MCA",semester:3,city:"Rajkot",mark:92}
                       ])

db.students.find()

db.students.findOne({name:"Rahul"})

db.students.find({city:"Rajkot"})

db.students.find({mark:{$gt:80}})

db.students.find({mark:{$lt:70}})

db.students.find({age:{$gte:23}})

db.students.find({semester:3})

db.students.find({},{name:1,mark:1,_id:0})

db.students.find({mark: {$gte: 70, $lte: 90}})

db.students.find({$or:[{city:"Rajkot"},{city:"Surat"}]})

db.students.find({ mark: { $gt: 80 }, semester:3});

db.students.find().sort({mark:-1}).limit(3)
```

```
db.students.countDocuments({})

db.students.find({name: /^R/})
```

Update Operation

```
db.students.updateOne({name:"Rahul"},{$set:{mark:95}})

db.students.updateOne({name:"Priya"},{$set:{city:"Mumbia"}})

db.students.updateMany({},{$inc:{mark:5}})

db.students.updateMany({},{$set:{status:"Active"}})

db.students.updateMany({},{$set:{status:"Active"}})

db.students.updateMany({},{$rename:{mark:"totalmark"}})

db.students.updateMany({},{$rename:{mark:"totalmark"}})

db.students.replaceOne({name:"Karan"},{rollno:105,age:22,course:"MCA",semester:2,city:"Surat",totalmark:80})

db.students.updateOne({name:"Riya"},{$set:{city:"Vadodara",semester:4}})

db.students.updateMany({},{$inc:{age:1}})
```

Delete Operation

```
db.students.deleteOne({name:"Amit"})

db.students.deleteOne({city:"Surat"})

db.students.deleteMany({semester:4})

db.students.deleteMany({totalmark:{$lt:75}})

db.students.deleteMany({city:"Rajkot"})

db.students.deleteMany({})

db.students.drop()
```

ASSIGNMENT EXERCISES

```
use LibraryDB


db.createCollection("books")


db.books.insertMany([
   {
      book_id: 101,
      title: "Mastering Java",
      author: "James Gosling",
      price: 650,
      category: "Programming",
      quantity: 10
   },
   {
      book_id: 102,
      title: "MongoDB Basics",
      author: "John Smith",
      price: 450,
      category: "Database",
      quantity: 8
   },
   {
      book_id: 103,
      title: "Python Programming",
      author: "Guido van Rossum",
      price: 750,
      category: "Programming",
      quantity: 4
   },
   {
      book_id: 104,
      title: "Computer Networks",
      author: "Andrew Tanenbaum",
      price: 600,
      category: "Networking",
      quantity: 6
   },
   {
      book_id: 105,
      title: "MySQL Guide",
      author: "Paul DuBois",
      price: 350,
      category: "Database",
      quantity: 3
   },
   {
      book_id: 106,
      title: "Modern C++",
      author: "Bjarne Stroustrup",
```

```
    price: 850,
    category: "Programming",
    quantity: 7
  },
  {
    book_id: 107,
    title: "Network Security",
    author: "Charlie Kaufman",
    price: 550,
    category: "Networking",
    quantity: 0
  },
  {
    book_id: 108,
    title: "Machine Learning",
    author: "Tom Mitchell",
    price: 700,
    category: "Programming",
    quantity: 5
  },
  {
    book_id: 109,
    title: "SQL Fundamentals",
    author: "John J. Patrick",
    price: 180,
    category: "Database",
    quantity: 12
  },
  {
    book_id: 110,
    title: "Web Development",
    author: "Jon Duckett",
    price: 280,
    category: "Programming",
    quantity: 2
  }
]);


db.books.find()


db.books.find({price:{$gt:500}})


db.books.find({category:"Programming"})


db.books.find({},{title:1,author:1,_id:0});


db.books.countDocuments()
```